

# **Engineering Mechanics**

## **Lecture 5: Equilibrium in 3-D**

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## EQUILIBRIUM IN THREE DIMENSIONS

- EQUILIBRIUM CONDITIONS

$$\Sigma \mathbf{F} = 0$$

$$\Sigma F_x = \Sigma \mathbf{F} \cdot \mathbf{i} = 0$$

$$\Sigma F_y = \Sigma \mathbf{F} \cdot \mathbf{j} = 0$$

$$\Sigma F_z = \Sigma \mathbf{F} \cdot \mathbf{k} = 0$$

$$\Sigma \mathbf{M} = 0$$

$$\Sigma M_x = \Sigma \mathbf{M} \cdot \mathbf{i} = 0$$

$$\Sigma M_y = \Sigma \mathbf{M} \cdot \mathbf{j} = 0$$

$$\Sigma M_z = \Sigma \mathbf{M} \cdot \mathbf{k} = 0$$

Moment due to forces can be obtained as  $\Sigma(\mathbf{r} \times \mathbf{F})$

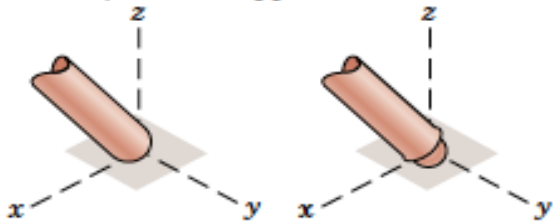
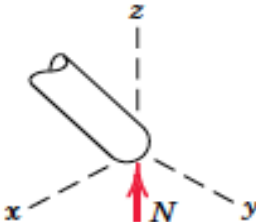
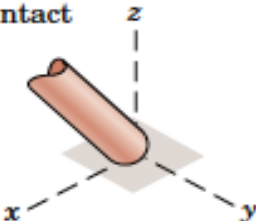
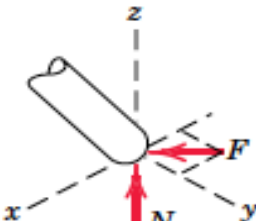
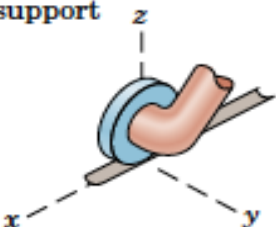
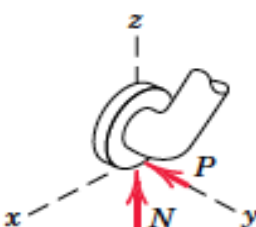
The six scalar relationships are independent conditions

**Caution:** Use Right hand frame

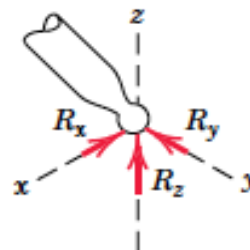
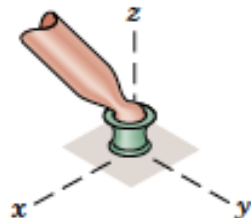
## Freebody Diagram

- The free-body diagram is the **most important single step** in the solution of problems in 3D.
- The essential purpose of the free-body diagram is **to develop a reliable picture of the physical action of all forces** (and couples if any) acting on a body.

# Action of Force in 3D

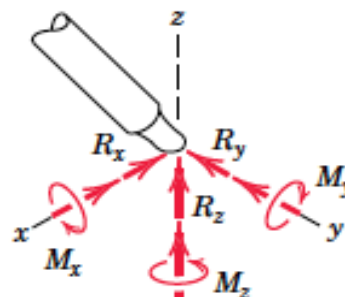
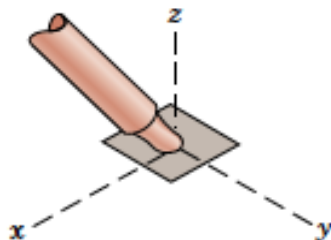
| MODELING THE ACTION OF FORCES IN THREE-DIMENSIONAL ANALYSIS                                                                                                 |                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type of Contact and Force Origin                                                                                                                            | Action on Body to Be Isolated                                                                                                                                                                                                                       |
| <p>1. Member in contact with smooth surface, or ball-supported member</p>  |  <p>Force must be normal to the surface and directed toward the member.</p>                                                                                       |
| <p>2. Member in contact with rough surface</p>                             |  <p>The possibility exists for a force <math>F</math> tangent to the surface (friction force) to act on the member, as well as a normal force <math>N</math>.</p> |
| <p>3. Roller or wheel support with lateral constraint</p>                |  <p>A lateral force <math>P</math> exerted by the guide on the wheel can exist, in addition to the normal force <math>N</math>.</p>                             |

#### 4. Ball-and-socket joint



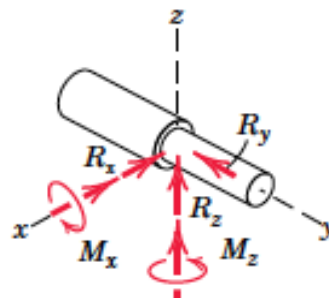
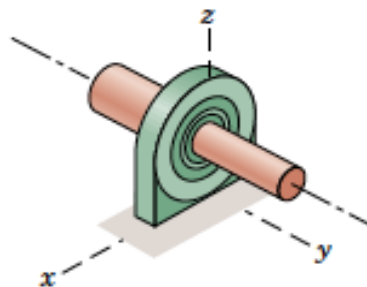
A ball-and-socket joint free to pivot about the center of the ball can support a force  $\mathbf{R}$  with all three components.

#### 5. Fixed connection (embedded or welded)



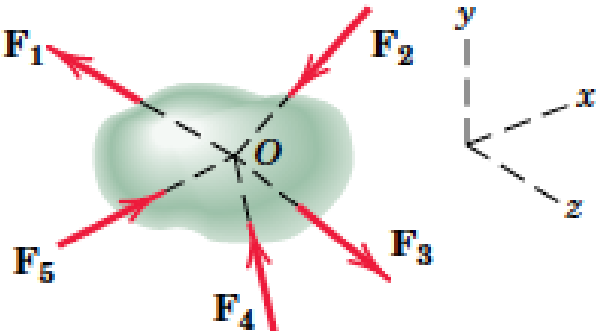
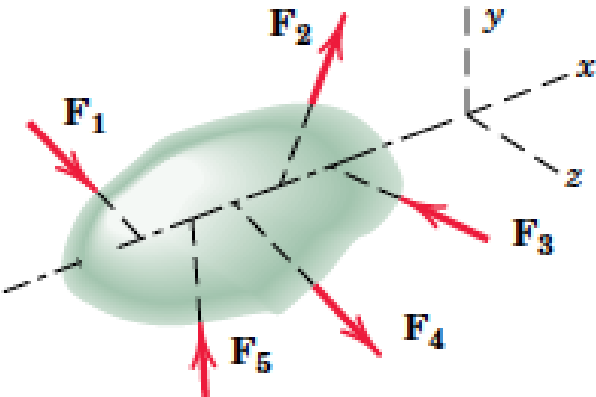
In addition to three components of force, a fixed connection can support a couple  $\mathbf{M}$  represented by its three components.

#### 6. Thrust-bearing support

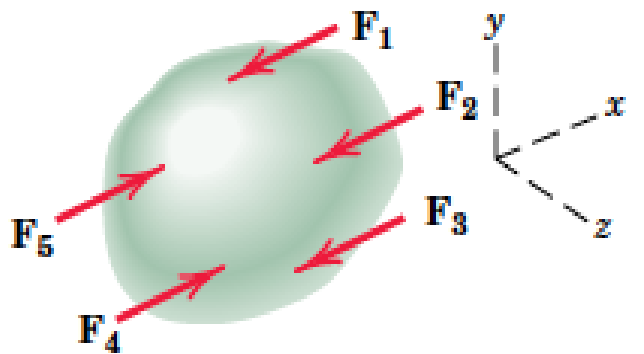
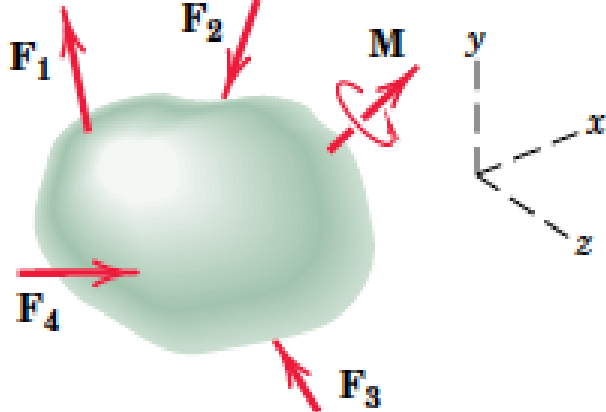


Thrust bearing is capable of supporting axial force  $R_y$  as well as radial forces  $R_x$  and  $R_z$ . Couples  $M_x$  and  $M_z$  must, in some cases, be assumed zero in order to provide statical determinacy.

# Categories of Equilibrium

| CATEGORIES OF EQUILIBRIUM IN THREE DIMENSIONS |                                                                                     |                                                                                              |
|-----------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Force System                                  | Free-Body Diagram                                                                   | Independent Equations                                                                        |
| 1. Concurrent at a point                      |   | $\Sigma F_x = 0$ $\Sigma F_y = 0$ $\Sigma F_z = 0$                                           |
| 2. Concurrent with a line                     |  | $\Sigma F_x = 0 \quad \Sigma M_y = 0$ $\Sigma F_y = 0 \quad \Sigma M_z = 0$ $\Sigma F_z = 0$ |

## Categories of Equilibrium

|             |                                                                                     |                                                                                                       |
|-------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 3. Parallel |   | $\Sigma F_x = 0$ $\Sigma M_y = 0$ $\Sigma M_z = 0$                                                    |
| 4. General  |  | $\Sigma F_x = 0$ $\Sigma F_y = 0$ $\Sigma F_z = 0$ $\Sigma M_x = 0$ $\Sigma M_y = 0$ $\Sigma M_z = 0$ |

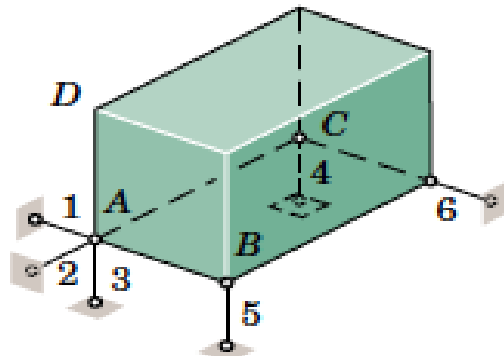
## Constraints and Statical Determinacy

- The **six scalar relations** necessary and sufficient conditions to establish equilibrium,
- **Do not necessarily provide** all of the information required to calculate the unknown forces acting **in a 3D equilibrium** situation



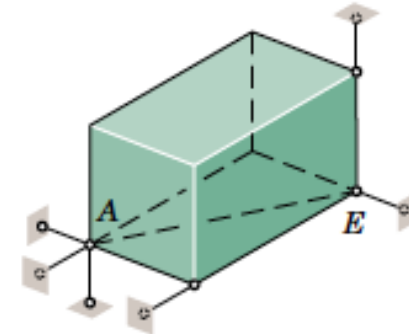
# Adequacy of Constraints

## Adequate Constraints



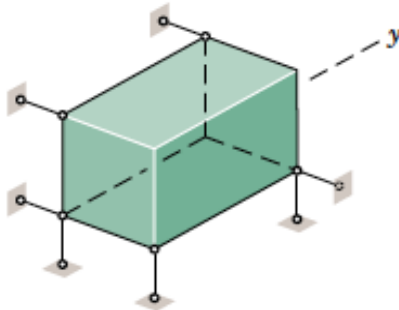
(a) Complete fixity  
Adequate constraints

## Partial Constraints: Can rotate about AE



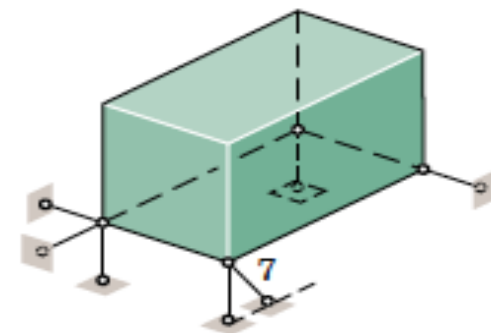
(b) Incomplete fixity  
Partial constraints

## Partial Constraints: Can translate along y



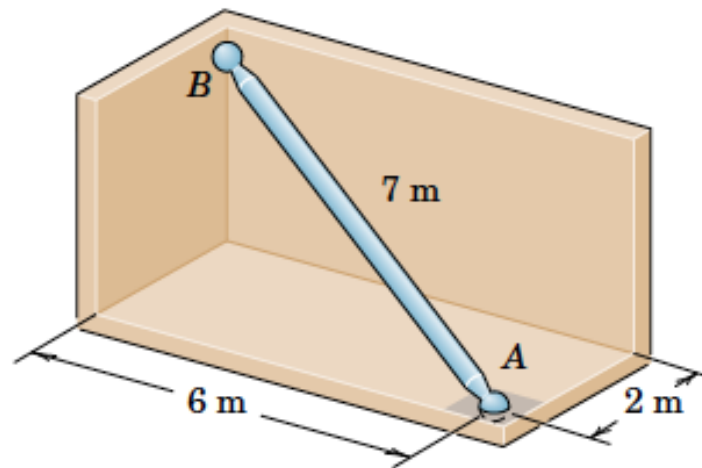
(c) Incomplete fixity  
Partial constraints

## Redundant Constraints

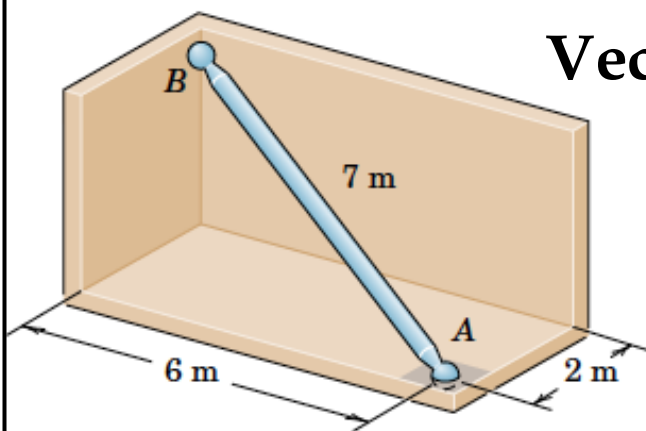


(d) Excessive fixity  
Redundant constraints

- Example: The uniform 7-m steel shaft has a mass of 200 kg and is supported by a ball and socket joint at A in the horizontal floor. The ball end B rests against the smooth vertical walls as shown. Compute the forces exerted by the walls and the floor on the ends of the shaft.



## Vector Approach



$$W = mg = 200(9.81) = 1962 \text{ N}$$

$$7 = \sqrt{2^2 + 6^2 + h^2},$$

$$h = 3 \text{ m}$$

$$\mathbf{r}_{AB} = -2\mathbf{i} - 6\mathbf{j} + 3\mathbf{k} \text{ m}$$

$$\mathbf{r}_{AG} = -1\mathbf{i} - 3\mathbf{j} + 1.5\mathbf{k} \text{ m}$$

$$[\Sigma \mathbf{M}_A = 0]$$

$$\mathbf{r}_{AB} \times (\mathbf{B}_x + \mathbf{B}_y) + \mathbf{r}_{AG} \times \mathbf{W} = 0$$

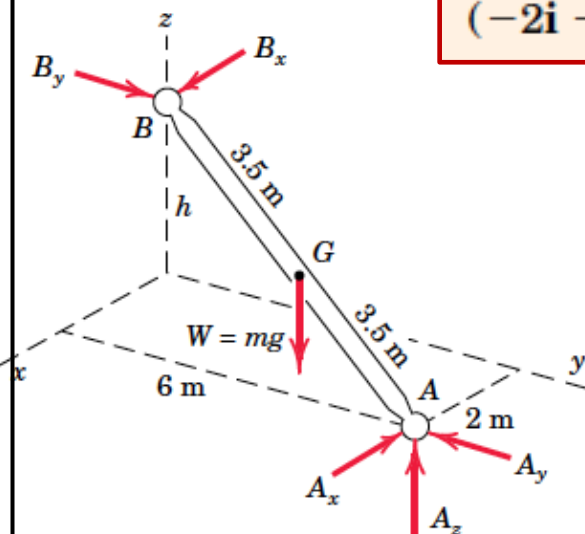
$$(-2\mathbf{i} - 6\mathbf{j} + 3\mathbf{k}) \times (B_x\mathbf{i} + B_y\mathbf{j}) + (-1\mathbf{i} - 3\mathbf{j} + 1.5\mathbf{k}) \times (-1962\mathbf{k}) = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & -6 & 3 \\ B_x & B_y & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -3 & 1.5 \\ 0 & 0 & -1962 \end{vmatrix} = 0$$

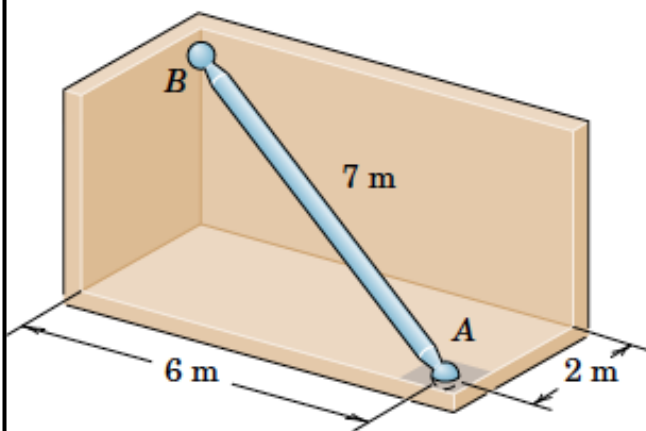
$$(-3B_y + 5890)\mathbf{i} + (3B_x - 1962)\mathbf{j} + (-2B_y + 6B_x)\mathbf{k} = 0$$

$$B_x = 654 \text{ N}$$

$$B_y = 1962 \text{ N}$$



## Vector Approach



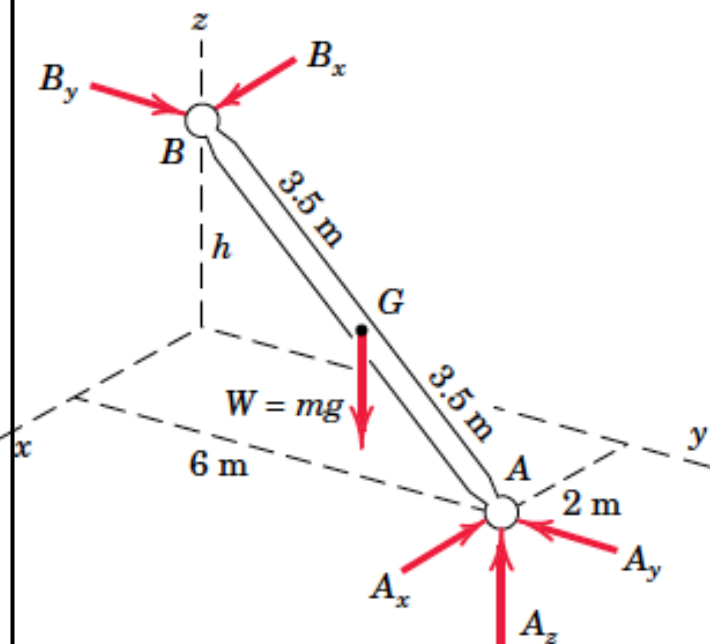
$$[\Sigma \mathbf{F} = 0]$$

$$(654 - A_x)\mathbf{i} + (1962 - A_y)\mathbf{j} + (-1962 + A_z)\mathbf{k} = 0$$

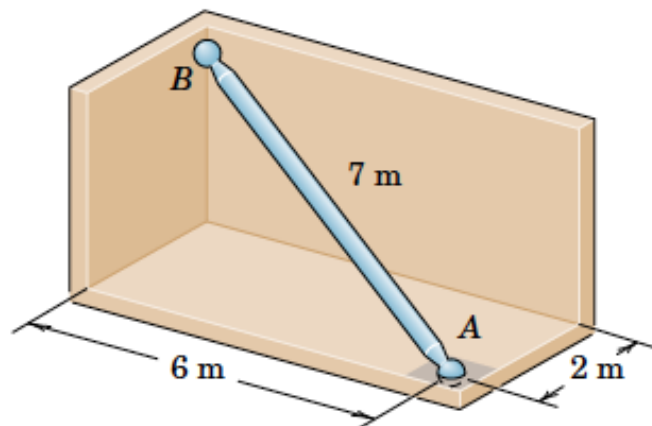
$$A_x = 654 \text{ N} \quad A_y = 1962 \text{ N} \quad A_z = 1962 \text{ N}$$

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$= \sqrt{(654)^2 + (1962)^2 + (1962)^2} = 2850 \text{ N}$$



## Scalar Approach



$$[\Sigma M_{A_x} = 0]$$

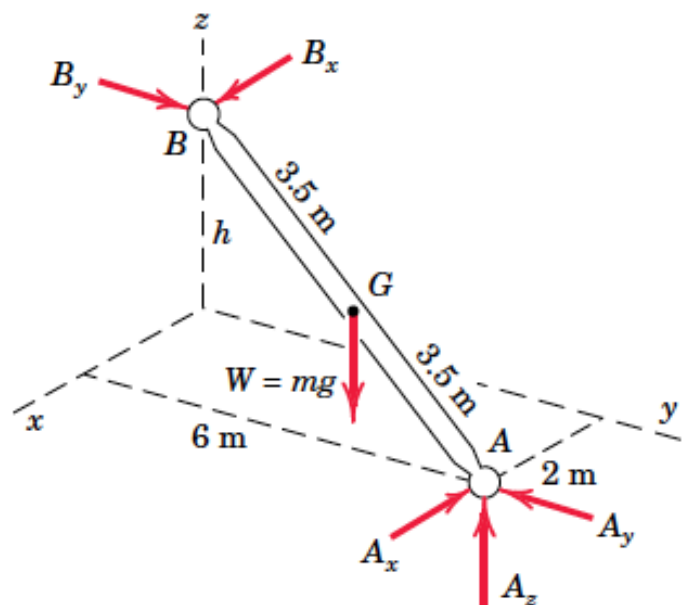
$$1962(3) - 3B_y = 0$$

$$B_y = 1962 \text{ N}$$

$$[\Sigma M_{A_y} = 0]$$

$$-1962(1) + 3B_x = 0$$

$$B_x = 654 \text{ N}$$



$$[\Sigma F_x = 0]$$

$$-A_x + 654 = 0$$

$$A_x = 654 \text{ N}$$

$$[\Sigma F_y = 0]$$

$$-A_y + 1962 = 0$$

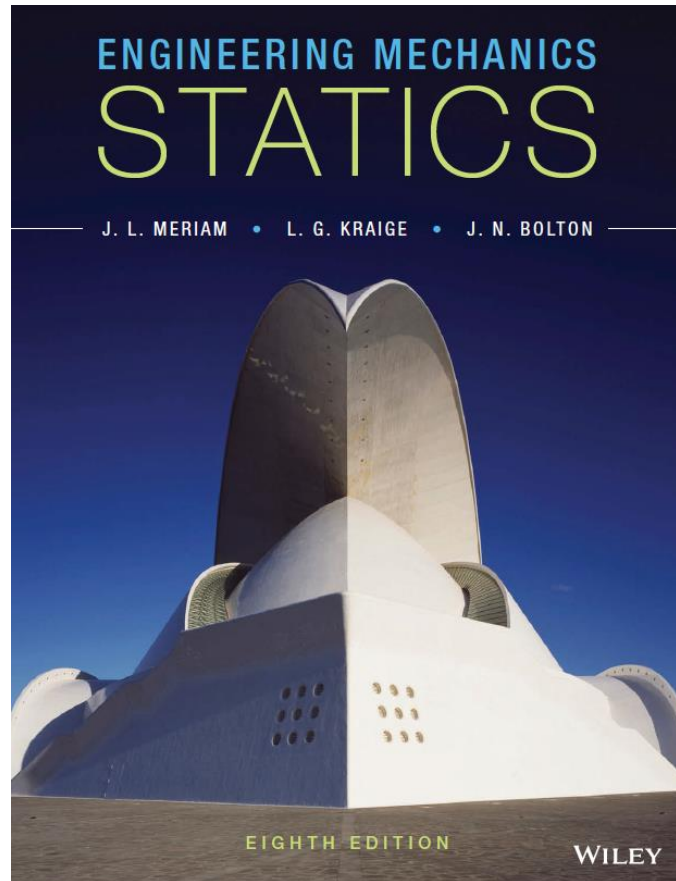
$$A_y = 1962 \text{ N}$$

$$[\Sigma F_z = 0]$$

$$A_z - 1962 = 0$$

$$A_z = 1962 \text{ N}$$

## References :



# THANK YOU